

APPENDIX 10 – ANALYSIS OF OUTCOME GROUPS

Updating Outcome Categories to Define Their Ranges

So far in this Chapter, the research has mapped the terrain of conditions: the design choices, organizational features, resources, and contextual factors that community-led digital stewardship initiatives bring to their work. After positioning cases within those condition ranges, revealing organizational signatures and common configurations, the next step is to turn to outcomes and ask *what do these configurations actually produce?*

Here are the four groups of outcomes, based on the original master tables prepared for the qualitative content analysis (Appendix 2). Democratic Practices (65 rows, 9 categories) addresses RQ I: whether and how community-led stewardship of digital resources enables democratic practices. Community Value (45 rows, 5 categories) addresses RQ II: how such stewardship generates community value, in multiple forms. Balance (38 rows, 5 categories) addresses RQ III: how initiatives navigate the inherent tensions in the work they do. Impact (36 rows, 6 categories) extends all three questions by asking what happens when community-led alternatives interact with, and transform, the broader systems in which they operate.

These outcome groups are interlocking dimensions of a coherent phenomenon. The five cases achieving outcomes across all four groups (Guifi.net, Masakhane, Matrix, SOLshare, and Te Hiku Media) demonstrate this. The analysis that follows examines each group's internal range before exploring the connections that bind them together.

Democratic practices

Democratic Practices captures the concrete manifestations of democratic values. The 65 outcome rows span from internal governance structures (how decisions get made within initiatives), to external control mechanisms (how communities exercise authority over data, infrastructure, and systems that affect them), to the cultural and legal frameworks that protect democratic arrangements over time.

How outcomes manifest

Mainly through observable structures and documented processes: governance bylaws establishing rotating leadership, data governance frameworks specifying consent mechanisms, infrastructure ownership arrangements, participation metrics showing who is involved in decisions, and legal protections codifying community control. They are measurable as well. Does the cooperative actually have member voting? Does the data framework actually give data subjects authority? Does the infrastructure actually remain under community control when external pressures mount?

Democratic Practices as a group generates the highest total condition connections (258), suggesting that multiple pathways lead to democratic governance. The most connected outcome is DP1 (Decentralized Organizational Governance, resonance=42), followed by DP2 (Community Data Sovereignty, resonance=38) and DP5 (Open Access and Transparent Systems, resonance=35). The least connected is DP9 (Local Capacity and Self-Governance, resonance=13), indicating this outcome requires more specialized conditions.

Significance

Democratic Practices directly addresses RQ I. The research question does not assume democracy emerges automatically from community ownership or alternative organizational forms. Democracy is not treated as a presumed consequence of good intentions.

The theoretical stakes are substantial. If community-led digital initiatives cannot demonstrate democratic practices, the case for alternatives to corporate or state control weakens. But the 65 outcome rows across 43 cases are empirical evidence that democratic practices do emerge, through multiple pathways and in multiple forms. The practical stakes are equally significant: these outcomes represent templates, proof points, and precedents that other initiatives can learn from and adapt.

The range of democratic practices

The nine categories within Democratic Practices create a spectrum from internal organizational democracy to external systemic transformation, with intermediate positions addressing different dimensions of democratic life.

Dimension 1: Locus of Democratic Practice

Some outcomes establish democracy within organizations (DP1: Decentralized Organizational Governance), while others extend democratic authority outward to data (DP2), infrastructure (DP3), and external systems (DP8: Resistance and Accountability Mechanisms). Democratic practice encompasses the full scope of an initiative's engagement with the world.

Dimension 2: Material vs. Procedural Foundations

Categories DP2, DP3, and DP9 address material foundations: control over data, infrastructure, and technical capacity without which democratic governance remains abstract. Categories DP1, DP4, DP5, and DP8 address procedural dimensions: the decision-making processes, participation structures, transparency mechanisms, and accountability arrangements that make governance actually democratic. Categories DP6 and DP7 bridge both through cultural and legal frameworks that shape what democratic practice means and protect it over time.

Dimension 3: Universal vs. Culturally-Specific Forms

DP6 (Indigenous and Cultural Governance Integration) shows how democratic practice takes culturally-specific forms. Māori data governance (Te Mana Raraunga, Te Hiku Media), First Nations OCAP principles (FNIGC), and Indigenous collective authority (GIDA) represent democratic practices grounded in non-Western frameworks. These outcomes challenge assumptions that democracy means one thing and demonstrate the pluriversal character of collective self-governance.

Overall, there are nine categories of democratic practices.

DP1: Decentralized Organizational Governance (10 cases, resonance=42)

This category captures outcomes that distribute decision-making authority within organizations, shifting power from hierarchy to collective process. The unifying element is restructured authority: rotating coordination (Loomio), federated structures (CoopCycle, Guifi.net), deliberation-based decision-making (The Drivers Cooperative), participatory policy development (Participatory Brazil), stigmergic contribution tracking (Sensorica), distributed research autonomy (DAIR), community-driven collaboration (Detroit CTC), self-managed networks (Enspiral), and participatory maker spaces (Nea Guinea/Tzoumakers).

Achieving DP1 outcomes indicates an organization has operationalized non-hierarchical governance. GV1 (Decentralized Authority Distribution) is the primary enabling condition (6 connections), confirming the intuitive relationship between distributing authority as a design choice and achieving distributed governance as an outcome.

DP2: Community Data Sovereignty and Control (8 cases, resonance=38)

This category includes outcomes establishing community, citizen, or Indigenous authority over data governance. The contexts vary¹, but the mechanism is consistent: data subjects exercise governance authority over information concerning them.

Achieving DP2 outcomes suggests that an organization has restructured data relations, moving from extraction (data as asset taken from subjects) to sovereignty (data as resource governed by subjects). Condition TP1 (Democratic Data Authority) shows the strongest condition connection (6), representing a near-definitional relationship: practices that establish data authority produce data sovereignty outcomes.

DP3: Community-Controlled Infrastructure (5 cases, resonance=22)

¹ Indigenous data sovereignty (FNIGC, GIDA, MAIAM NAYRI WINGARA), citizen health data (MiData, OpenSAFELY), worker data (WAO), municipal data (DECODE), personal communications (Matrix).

This category tackles community ownership and control over physical and digital infrastructure. These represent the material foundation for other democratic practices; without control over essential infrastructure, governance remains dependent on external actors.

Achieving DP3 outcomes implies an organization has created durable material assets under community control. TT1 (Community-Controlled Physical Infrastructure) shows the strongest condition connection (5) and 100% coverage; it is a necessary condition.

DP4: Inclusive Participation and Representation (6 cases, resonance=33)

This category features outcomes that enable marginalized groups to participate in and lead decision-making: centering marginalized knowledge (CLEAR), leadership roles for marginalized groups (DEF), gender equity in workplace governance (Les Mercedes), integration of technical expertise with community linguistic knowledge (Masakhane), broad citizen input (Participatory Brazil), and directly-affected community leadership (Stop LAPD Spying Coalition).

Achieving DP4 outcomes means an organization has operationalized inclusion. SC1 (Inclusive Participation Structures) shows the strongest condition connection (4), suggesting that intentional design for inclusion produces inclusive outcomes.

DP5: Open Access and Transparent Systems (7 cases, resonance=35)

This category is comprised of outcomes creating systems accessible to all: open collaborative development (Linux Foundation), accessible decision-making tools (Loomio), community control over communication infrastructure (Matrix), transparent technical processes (Open Mobility Foundation), open mapping commons (OpenStreetMap), citizen consultation platforms (Participatory Brazil), and transparent pricing with privacy respect (The Drivers Cooperative).

Achieving DP5 outcomes signifies that an organization has made efforts to render its systems accessible and transparent, which are preconditions for meaningful democratic participation. Condition TP2 (Open Development and Transparency) shows 4 connections, and TT4 (Participatory Design Platforms) shows 3, confirming that open practices and participatory tools produce open, transparent outcomes.

DP6: Indigenous and Cultural Governance Integration (4 cases, resonance=24)

This category portrays outcomes embedding Indigenous or culturally-specific governance frameworks in technology and data management: First Nations values in data governance (FNIGC), global Indigenous collective governance (GIDA), Māori cultural governance in AI and archives (Te Hiku Media), and culturally-grounded Māori data sovereignty (Te Mana Raraunga).

Achieving DP6 outcomes indicates an organization has operationalized culturally-specific forms of collective governance. Condition PU1 (Indigenous Epistemological Centering) shows 4 connections with 100% coverage.

DP7: Legal and Institutional Protection (6 cases, resonance=33)

This category describes outcomes establishing legal frameworks protecting community control: legal protection for cooperative technology (CoopCycle), strategic partnerships advancing Indigenous governance (FNIGC), legal protection for commons-based telecommunications (Guifi.net), community trusts holding assets (Solar Commons), legal frameworks protecting Māori data sovereignty (Te Hiku Media), and legal victories establishing Indigenous telecommunications autonomy (TIC A.C.).

Achieving DP7 outcomes means an organization has institutionalized protection for democratic practices, making community control durable against appropriation or dissolution. Condition RG1 (Legal Innovation for Sovereignty Protection) shows 4

connections with 100% coverage; legal protection outcomes require legal innovation conditions.

DP8: Resistance and Accountability Mechanisms (4 cases, resonance=18)

This category is a collection of outcomes challenging harmful systems and creating accountability: integrated approach building alternatives while challenging harm (CLEAR), community-driven AI reducing bias (DAIR), shutdown of surveillance programs (Stop LAPD Spying Coalition), and worker-controlled algorithmic investigation (WAO).

Achieving DP8 outcomes implies an organization has developed the capacity to contest external power. This is the only Democratic Practices category where no condition shows 3+ connections, suggesting resistance outcomes emerge from diverse condition combinations.

DP9: Local Capacity and Self-Governance (3 cases, resonance=13)

These outcomes build local technical capacity for self-governance: local digital resource governance (Detroit CTC), digital-physical integration for citizen participation (Participatory Brazil), and local technical capacity for independent network maintenance (Zenzeleni).

DP9 outcomes paint the picture of organizations that have created capability for ongoing self-governance. The lower resonance (13) suggests this outcome requires more specialized condition configurations than other Democratic Practices categories.

Intra-Group connections

Democratic Practices categories interconnect through three pathways:

- **Foundation → Practice → Protection:** Material foundations (DP3: Infrastructure, DP9: Capacity) enable democratic practices (DP1: Governance, DP4: Inclusion, DP5: Transparency), which in turn require protection (DP7: Legal) to endure. Guifi.net exemplifies this pathway: community-controlled telecommunications (DP3) enables federated governance (DP1), protected by legal innovation (DP7).
- **Authority → Accountability:** Data sovereignty (DP2) and resistance mechanisms (DP8) connect through the exercise of authority over systems. WAO

demonstrates both: worker-controlled data tools (DP2) enable algorithmic investigation and accountability (DP8).

- **Universal → Culturally-Specific:** Decentralized governance (DP1) and cultural governance integration (DP6) represent different expressions of collective self-governance. FNIGC demonstrates both: distributed authority within (DP1) grounded in First Nations values (DP6).

Community Value

These are outcomes that generate collective wealth, resources, or benefits at the community level. The “value” is that which accrues to communities instead of being extracted. The 45 outcome rows span from economic value (worker-retained earnings, cooperative profits, community-circulating revenue), to knowledge commons (open-source tools, research datasets, accessible platforms), to cultural value (linguistic preservation, heritage technology, culturally-embedded resources). This is value in all its facets, unbounded by monetary terms.

How outcomes manifest

Community Value outcomes are observed through documented flows and tangible assets: revenue retained by workers, open-source code accessible to all, community infrastructure generating ongoing benefits, language technology preserving endangered tongues, and alternative economic arrangements that restructure relationships between work, ownership, and returns.

The resonance analysis shows Community Value with total connections of 135. The most connected outcome is CV2 (Open Knowledge and Technology Commons, resonance=35), reflecting the generative power of open production. CV5 (Alternative Economic Models and Structures, resonance=17) shows lower connectivity, suggesting that structural economic innovation requires more specialized conditions.

Significance

Community Value directly addresses RQ II: how community-led stewardship of digital resources generates value that accrues to communities. This matters theoretically because it challenges narrow economic conceptions of value. If community wealth only means monetary profit, initiatives creating cultural preservation, knowledge commons, and capacity assets would be invisible. But the 45 outcome rows establish that community value takes multiple forms (economic, epistemic, cultural, infrastructural) each representing genuine wealth creation.

It matters practically because community-led initiatives operate in resource-constrained environments. Value creation provides evidence to counter claims that alternatives to the dominant digital paradigm are economically unviable.

The range of community value

Value creation operates through distinct mechanisms.

Dimension 1: Retention vs. Creation

Some outcomes retain value that would otherwise be extracted: CV1 (Worker and Member Economic Benefits) captures value that in conventional arrangements flows to shareholders or platform owners. Others create new forms of value: CV2 (Open Knowledge and Technology Commons) generates wealth through commons production that didn't exist before. CV3 (Community Infrastructure and Capacity Assets) operates both ways, building new assets while capturing value that external infrastructure providers would extract.

Dimension 2: Economic vs. Non-Economic Value

CV1 and CV5 concern economic value directly: wages, profits, and financial arrangements. CV2 and CV4 concern knowledge and cultural value (open resources, linguistic preservation, heritage technology). CV3 bridges both through infrastructure that

generates economic returns while creating non-monetary community assets. This dimension means that "community value" encompasses more than money.

Dimension 3: Individual vs. Collective Beneficiary

CV1 delivers benefits to individual workers and members through ownership structures. CV2 creates commons accessible to all. CV4 embeds value in culturally-specific forms benefiting particular communities. The range shows that community value flows to communities through different pathways.

Overall, there are five categories of community value.

CV1: Worker and Member Economic Benefits (7 cases, resonance=28)

This category includes outcomes delivering direct economic benefits through ownership structures that retain value within communities: worker-owned alternatives to exploitative platforms (CoopCycle), worker-retained value through cooperative ownership (CoopCycle), community-selected projects funded through redistributed commissions (Fairbnb.coop), tourism revenues redistributed locally (Fairbnb.coop), cultural value retained in community (Groupmuse), fair wages with social protections (Les Mercedes), and cooperative structures retaining earnings (The Drivers Cooperative).

CV1 outcomes imply an organization has structured ownership to benefit workers and members directly. Condition GV2 (Ownership-Based Democratic Control) shows 5 connections; the ownership form that enables democratic governance also produces worker economic benefits. RS3 (Economic Value Redistribution) shows 4 connections, confirming that redistribution conditions produce redistribution outcomes.

CV2: Open Knowledge and Technology Commons (9 cases, resonance=35)

This category comprises outcomes creating freely accessible knowledge, tools, and technology: open-source data sovereignty tools (DECODE), collaborative governance ensuring public access (Linux Foundation), free deliberation tools (Loomio), open

communication protocols (Matrix), free geographic data (OpenStreetMap), global mapping databases (P2P Foundation), comprehensive commons ecosystems (P2P Foundation), flexible research data standards (RDA), open collaborative research (Sensorica), and open-source construction systems (WikiHouse).

Achieving CV2 outcomes indicates an organization is making resources accessible to all without enclosure. Condition TP2 (Open Development and Transparency) shows 7 connections, the highest for any condition-outcome pair in the Community Value group. It confirms that open development practices produce open commons outcomes.

CV3: Community Infrastructure and Capacity Assets (6 cases, resonance=28)

This category portrays outcomes creating durable community-controlled assets: diversified funding for national research data assets (ARDC), portable community networking (Detroit CTC), community technological capacity building (DEF), 40,000+ node commons-based telecommunications (Guifi.net), peer-to-peer electricity trading networks (SOLshare), and solar benefits generating long-term wealth (Solar Commons).

Achieving CV3 outcomes indicates an organization has built durable capital at the community level; infrastructure and capacity that generate ongoing value beyond immediate transactions. Conditions TP3 (Multi-Scale Architecture Design) and TT1 (Physical Infrastructure) each show 3 connections, suggesting that technical architecture and physical infrastructure conditions produce community asset outcomes.

CV4: Culturally-Specific Technology and Knowledge (6 cases, resonance=27)

This category consists of outcomes embedding technology or knowledge in specific cultural contexts: local technological literacy (DEF), community music events creating cultural value (Groupmuse), Indigenous data valued as collective asset (MAIAM NAYRI WINGARA), open-access datasets for African languages (Masakhane), Tibetan NLP tools

integrated with monastic institutions (Monlam AI), and Māori language transcription exceeding commercial models (Te Hiku Media).

Producing CV4 outcomes denotes an organization that has created value that is irreducibly cultural. TT2 (Culturally-Grounded Technology Artifacts) and PU2 (Cultural Preservation Purpose) each show 3 connections, confirming that cultural conditions produce cultural value outcomes.

CV5: Alternative Economic Models and Structures (4 cases, resonance=17)

This category covers outcomes creating novel economic arrangements: accessible research tools democratizing inquiry (CLEAR), participatory resource allocation across networked enterprises (Enspiral), novel organizational structure enabling commons-based peer production (Sensorica), and hybrid governance combining direct worker involvement with expert analysis (WAO).

Achieving CV5 outcomes suggests an organization is working to reimagine economic relations. The lower resonance (17) suggests these outcomes require more specific condition configurations; no single condition shows dominant connections.

Intra-Group connections

Community Value categories connect through complementary value forms:

- **Retention → Commons → Innovation:** Ownership-based retention (CV1) creates the material stability for commons production (CV2), which in turn enables structural innovation (CV5). CoopCycle demonstrates this pathway: cooperative ownership retains worker value (CV1), the technology remains open for other cooperatives (implied CV2 connection), enabling the federated cooperative model to spread.
- **Infrastructure → Capacity → Culture:** Infrastructure assets (CV3) create the foundation for capacity building, which enables culturally-specific development (CV4). DEF typifies this: village-owned infrastructure (CV3) enables digital literacy building, which supports local language applications (CV4).
- **Economic → Cultural Integration:** CV1 (economic benefits) and CV4 (cultural value) are integrated. Groupmuse showcases both: the cooperative structure retains cultural value economically (CV1) while the house concerts create cultural and social value that cannot be reduced to economics (CV4).

Balancing priorities for claimants of digital sovereignty

This outcome group is about cases engaging in broader systems. This is not balance as compromise or splitting the difference, but as dynamic navigation: achieving both local control and systemic participation, both community protection and external engagement, both particular needs and interoperable standards.

How outcomes manifest

Balance outcomes are enabled through governance architectures, partnership agreements, exchange protocols, and standards frameworks that structure relationships between local and trans-local actors. They are observable in federated structures that preserve node autonomy, in network positioning that amplifies voice without sacrificing identity, in bidirectional flows that ensure mutual benefit, and in flexible standards that enable both coordination and adaptation.

The resonance analysis shows Balance with a total number of connections of 143, between Community Value and Impact. The most connected outcome is BL4 (Community Sovereignty Protection in External Engagement, resonance=37), reflecting the centrality of sovereignty maintenance. The least connected is BL5 (Global-Local Standards Navigation, resonance=16), suggesting that standards navigation requires more specialized configurations.

Significance

Balance directly addresses RQ III: how initiatives navigate the tensions they face in performing their work. The research question recognizes that community-led alternatives cannot operate in isolation. They require resources, knowledge, legitimacy, and coordination that come from engagement with broader systems, but such engagement risks compromising the community control that defines them.

This is theoretically significant because it challenges both romantic notions of localism (the belief that community sovereignty means isolation) and naive cosmopolitanism (the belief that global coordination automatically benefits local actors). The 38 outcome rows indicate that sovereignty and engagement are not zero-sum. Initiatives can achieve federated governance that preserves local control while enabling systemic participation.

It matters practically because every community-led initiative faces partnership offers, standards pressures, and coordination opportunities that could enhance or undermine their work. Understanding how balance outcomes are achieved provides guidance for navigating these tensions without sacrificing either autonomy or effectiveness.

The range of balance

The cases have different mechanisms for achieving balance, along three main dimensions.

Dimension 1: Structural vs. Relational Solutions

BL1 (Federated Multi-Scale Governance) represents structural solutions, namely governance architectures that embed balance in organizational design. BL2 (Multi-Network Coordination) and BL3 (Bidirectional Exchange) represent relational solutions: patterns of engagement that achieve balance through reciprocity and positioning. BL4 (Sovereignty Protection) is boundary-setting: explicit protection of community control in partnerships. BL5 (Standards Navigation) consists of technical solutions, such as interoperability frameworks that enable both coordination and adaptation.

Dimension 2: Scale of Tension Addressed

All categories address scale-related tensions, but at different levels. BL1 concerns internal governance across scales (how nodes relate to networks). BL2 concerns external positioning across networks (how initiatives relate to movements and institutions). BL3

concerns knowledge and resource flows between partners. BL4 concerns power dynamics in external engagement. BL5 concerns technical standards and their adaptation to context.

Dimension 3: Proactive vs. Protective Orientation

BL1 and BL2 are proactively constructive, building architectures and positions that achieve balance. BL3 structures exchanges that could otherwise be extractive. BL4 is explicitly protective, maintaining sovereignty against partnership pressures. BL5 navigates between external standards and local needs.

There are five categories in the range of balance outcomes.

BL1: Federated Multi-Scale Governance (6 cases, resonance=30)

This category captures outcomes establishing federated structures that enable both local autonomy and systemic coordination: federated structures combining autonomy with shared principles (Guifi.net), bounded flexibility ensuring interoperability with adaptation (Guifi.net), federated architecture combining sovereignty with interoperability (Matrix), multi-scale architecture for diverse contexts (Matrix), interconnected cooperatives enabling cross-border research while preserving local control (MiData), city-led governance with interoperable standards (Open Mobility Foundation), and farmer-controlled technology with global interoperability (OpenAgriNet, SOLshare).

Achieving BL1 outcomes indicates an organization has architecturally resolved the local-global tension through structural design enabling both. Condition TP3 (Multi-Scale Architecture Design) shows 5 connections and GV3 (Federated Multi-Scale Coordination) shows 3, confirming that multi-scale design conditions could produce multi-scale governance outcomes.

BL2: Multi-Network Coordination and Solidarity (6 cases, resonance=29)

This category is comprised of outcomes positioning initiatives across multiple networks for amplified voice and collective power: collaborative networks amplifying social

enterprise impact (Enspiral), transnational Indigenous coordination amplifying advocacy (GIDA), networks of cities collectively shaping mobility governance (Open Mobility Foundation), transnational knowledge infrastructure supporting commons movement (P2P Foundation), transnational network amplifying community-driven urban advocacy (SDI), and transnational Indigenous coordination amplifying Māori advocacy (Te Mana Raraunga).

Achieving BL2 outcomes means an organization has strategically positioned itself to build collective power without sacrificing organizational identity. Condition PT2 (Multi-Network Strategic Positioning) shows 6 connections with 100% coverage. Network coordination outcomes require network positioning conditions.

BL3: Bidirectional Knowledge and Resource Exchange (7 cases, resonance=31)

This category depicts outcomes establishing reciprocal flows between partners: national research data infrastructure enabling collaboration (ARDC), multi-partner ecosystems enabling citizen-controlled governance (DECODE), reciprocal exchange strengthening both local and global initiatives (DEF), university partnerships ensuring community benefits from research (Masakhane), pro bono institutional support through collaboration (OpenSAFELY), reciprocal knowledge flows benefiting both global and local (P2P Foundation), and global data sharing infrastructure supporting research commons (RDA).

Achieving BL3 outcomes point to organizations that have structured partnerships for mutual benefit. Condition PT1 (Reciprocal Knowledge Exchange) shows 6 connections with 100% coverage.

BL4: Community Sovereignty Protection in External Engagement (9 cases, resonance=37)

This category portrays outcomes maintaining community control while engaging externally: locally-responsive growth maintaining partnerships (Fairbnb.coop), locally-rooted

alternatives addressing neighborhood needs (Les Mercedes), balanced engagement leveraging resources while preserving control (Linux Foundation), technology aligned with cultural values (Monlam AI), digital tools integrating traditional and modern knowledge (Nea Guinea/Tzoumakers), city-controlled mobility data standards (Open Mobility Foundation), community-led development with transnational solidarity (SDI), independent technology development retaining control (Te Hiku Media), and community assemblies determining priorities through consensus (TIC A.C.).

Achieving BL4 outcomes shows an organization has navigated partnership dynamics without ceding sovereignty. Conditions PT3 (Community Sovereignty Protection in Partnerships) and TT2 (Culturally-Grounded Technology Artifacts) each show 3 connections, suggesting that both relational boundary-setting and cultural grounding of technology contribute to sovereignty protection.

BL5: Global-Local Standards Navigation (4 cases, resonance=16)

This category refers to outcomes navigating tensions between global standards and local needs: multi-scale research data governance (ARDC), context-specific privacy standards within global dialogue (OpenSAFELY), local mapping autonomy within global collaborative commons (OpenStreetMap), and crowd-sourced innovation preserving open access (Sensorica).

If an organization has found practical accommodation between interoperability requirements and contextual specificity, they are working towards this outcome category. The lower resonance suggests this outcome requires more specialized condition configurations; no single condition dominates the pathway.

Intra-Group connections

Balance categories connect through nested scales of navigation:

- **Architecture → Positioning → Exchange:** Federated structures (BL1) create the governance framework within which network positioning (BL2) and reciprocal

exchange (BL3) occur. Guifi.net demonstrates this: federated governance (BL1) enables positioning across telecommunications and commons networks, which enables bidirectional knowledge sharing with similar initiatives globally.

- **Proactive → Protective:** Building balanced structures (BL1, BL2, BL3) creates the foundation from which sovereignty protection (BL4) operates. Without proactive construction of balanced arrangements, protection becomes purely defensive. Te Hiku Media exemplifies this: building independent technology development (proactive) enables successful protection of Māori data sovereignty in partnerships (protective).
- **Standards as Enabling Constraint:** BL5 (Standards Navigation) connects to all other categories by addressing the technical layer where balance must be achieved. Interoperable standards (BL5) enable federated governance (BL1), network coordination (BL2), and knowledge exchange (BL3) while requiring protection of local adaptation capacity (BL4).

Impact

Outcomes achieved by the 43 cases extend beyond organizational boundaries, including changes in policy, institutional frameworks, community capacity, cultural systems, economic arrangements, and ecological conditions. These changes in turn transform the contexts in which community-led initiatives operate.

Viewing the impact group as extending the effects of the previous three, impact is closely associated with the concept of digital sovereignty developed in this dissertation. Impact is evidence that a capacity is being constructed: changes in the rules governing systems, capacities that persist and multiply, cultural preservation that counters erasure, economic restructuring that demonstrates alternatives, and institutional reforms that create space for further change.

How outcomes manifest

Impact outcomes manifest through documented changes in external systems: policies adopted, regulations reformed, capacities built in communities beyond immediate participants, languages preserved, economic models replicated, governance institutions transformed, climate resilience enhanced. They are observable not just in organizational outputs but in systemic changes traceable to community-led initiative.

The resonance analysis reveals a striking pattern: Impact shows the highest single-outcome resonance of any group with IM1 (Policy and Regulatory Change) at 53 connections—more than any other outcome across all groups. This suggests that policy change serves as a multiplier, connected to conditions across all types. By contrast, IM6 (Climate and Ecological Transformation) shows only 11 connections, suggesting ecological impact requires more specialized pathways.

Significance

Impact extends all three research questions by asking what happens when the cases under study interact with and transform broader systems.

Connection to RQ I: Democratic Practices *become* Impact when they transform governance institutions beyond the initiative itself. Outcome IM5 (Governance and Institutional Transformation) captures this extension. Participatory Brazil's citizen participation platforms transform government, not just the initiative's internal governance. FNIGC's multi-scalar governance contributes to transnational Indigenous advocacy, extending democratic practice beyond organizational boundaries.

Connection to RQ II: Community Value *becomes* Impact when it restructures economic arrangements beyond the initiative. Outcome IM4 (Economic Value Transformation) demonstrates this extension. Guifi.net's community-retained value demonstrates viable alternatives to extractive telecommunications, influencing how other communities approach infrastructure development. Namma Yatri's zero-commission model challenges platform capitalism's fundamental premises.

Connection to RQ III: Balance *becomes* Impact when navigation strategies reshape the systems being navigated. Outcome IM1 (Policy and Regulatory Change) establishes this most directly. Initiatives that successfully navigate policy environments while advocating for change transform those environments for future initiatives. FNIGC's policy frameworks, Te

Mana Raraunga's formal agreement with Statistics New Zealand, and TIC A.C.'s regulatory exemption create more supportive conditions for community-led alternatives broadly.

Impact thus represents where community-led digital stewardship meets the world. It is where internal democratic practices, community value creation, and navigational balance extend outward to reshape the contexts in which all community-led initiatives must operate. The outcome group's 36 rows (from the 'master' table in Appendix 2) provide valuable and documented information about this extension: policies change, capacities build, cultures persist, economies restructure, institutions transform, and ecological conditions improve through community-led action.

The range of impact

Impact outcomes operate across three dimensions.

Dimension 1: Rules vs. Capacities vs. Conditions

IM1 (Policy and Regulatory Change) and IM5 (Governance and Institutional Transformation) change the rules; formal policies, regulations, and institutional frameworks. IM2 (Community Capacity Building) and IM4 (Economic Value Transformation) build capacities; ongoing capability for action and alternative economic models. IM3 (Cultural and Linguistic Sovereignty) and IM6 (Climate and Ecological Transformation) address conditions; cultural continuity and ecological resilience that enable community life.

Dimension 2: Direct Production vs. Demonstration Effects

Some impacts are directly produced by initiative action: policy victories (IM1), capacity built through training (IM2), languages preserved through technology (IM3). Others operate through demonstration: alternative economic models that show viability (IM4), governance innovations that others can learn from (IM5), climate solutions that can be replicated (IM6).

Dimension 3: Immediate vs. Generative

IM1 produces immediate policy change. IM2 builds capacity for future action through generative impact that multiplies over time. IM3 preserves cultural foundations for ongoing community life. IM4 creates economic alternatives that challenge dominant models. IM5 transforms institutions enabling further change. IM6 builds resilience for long-term sustainability.

These dimensions come into play when describing the six categories of impact.

IM1: Policy and Regulatory Change (10 cases, resonance=53)

This category features outcomes achieving changes in formal rules: community science informing policy (CLEAR), policy frameworks recognizing Indigenous data sovereignty (FNIGC), Indigenous governance frameworks adopted internationally (GIDA), open protocols shaping global communications (Matrix), public governance frameworks for mobility data (Open Mobility Foundation), regulatory frameworks returning control to public governance (Open Mobility Foundation), commons frameworks entering policy discussions (P2P Foundation), policy victories eliminating surveillance programs (Stop LAPD Spying Coalition), formal agreement recognizing Māori data sovereignty (Te Mana Raraunga), regulatory exemption enabling community telecommunications (TIC A.C.), and worker-owned models informing regulatory reform (The Drivers Cooperative).

Achieving IM1 outcomes places a case at the center of efforts to reshape the policy environment. PO1 (Policy and Systemic Transformation Advocacy) shows 9 connections (the highest single condition-outcome connection in the entire matrix) confirming that sustained advocacy produces policy change. The 100% coverage suggests advocacy is a necessary condition for policy impact.

IM2: Community Capacity Building (6 cases, resonance=35)

This category describes outcomes building ongoing capability in communities: enhanced capacity for contextually appropriate AI (DAIR), regional capacity for First

Nations data sovereignty (FNIGC), distributed African AI research capacity (Masakhane), enhanced citizen capacity for policy participation (Participatory Brazil), community-generated data empowering advocacy (SDI), and community capacity to challenge surveillance (Stop LAPD Spying Coalition).

Producing IM2 outcomes indicates an organization has created capacity that persists and multiplies beyond immediate outputs. Condition RS1 (Community Capacity Development) shows 6 connections with 100% coverage.

IM3: Cultural and Linguistic Sovereignty (3 cases, resonance=17)

This category depicts outcomes advancing cultural or linguistic preservation: African language technological capacity and methodology transformation (Masakhane), Tibetan linguistic sovereignty through community-controlled AI (Monlam AI), and world-leading Māori language AI advancing cultural revitalization (Te Hiku Media).

IM3 outcomes suggest an organization has contributed to cultural continuity against forces of erasure. Condition PU2 (Cultural Preservation Purpose) shows 3 connections, confirming that explicit cultural purpose orientation produces cultural sovereignty outcomes.

IM4: Economic Value Transformation (3 cases, resonance=15)

This category contains outcomes transforming economic arrangements: community-retained value through locally-circulating economy (Guifi.net), market-engaged alternative demonstrating viable non-extractive model (Namma Yatri), and transparent worker-controlled platform with fair distribution (The Drivers Cooperative).

IM4 outcomes propose that an organization has introduced alternative economic logics. Condition RS3 (Economic Value Redistribution) shows 3 connections, confirming that redistribution conditions produce transformation outcomes. The low case count (3) but clear mechanism suggests economic transformation remains an emerging frontier for community-led initiatives.

IM5: Governance and Institutional Transformation (6 cases, resonance=33)

This category defines outcomes transforming how governance institutions operate: community control over research (CLEAR), transnational growth with locally-responsive governance (CoopCycle), multi-scalar governance with local control and transnational advocacy (FNIGC), transparent privacy-preserving research with accountability (OpenSAFELY), government platforms for citizen participation (Participatory Brazil), and resident-led urban development recognized by governments (SDI).

IM5 outcomes indicates a case is influencing how institutions (including government) make decisions. Condition GV6 (Democratization of External Processes) and RS1 (Community Capacity Development) each show high connections, suggesting that institutional transformation requires both democratizing engagement with external systems and building capacity to sustain that engagement.

IM6: Climate and Ecological Transformation (3 cases, resonance=11)

This category is comprised of outcomes addressing climate and ecological challenges: community-led climate adaptation integrating housing justice and environmental resilience (SDI), community-controlled energy systems providing climate resilience (SOLshare), and integrated measurement capturing economic and social dimensions of energy transition (Som Energia).

Achieving IM6 outcomes suggests an organization has contributed to ecological objectives through community-controlled solutions. The low resonance (11) implies ecological impact requires specialized condition configurations. Conditions SC3 (Climate Crisis Response) and RS3 (Economic Value Redistribution) appear relevant but with limited connections. This may represent an underdeveloped impact pathway with potential for expansion.

Intra-Group connections

Impact categories connect through reinforcing pathways:

Capacity → Policy → Institution: Capacity building (IM2) creates the foundation for policy advocacy, which produces policy change (IM1), which enables institutional transformation (IM5). FNIGC embodies this complete pathway: regional capacity building (IM2) enables sustained advocacy, which produces national policy frameworks (IM1), which contributes to multi-scalar governance transformation (IM5).

Cultural → Economic Integration: Cultural sovereignty (IM3) and economic transformation (IM4) are not separate domains but integrated for some initiatives. Te Hiku Media's cultural preservation work has economic dimensions; Masakhane's language technology work builds research capacity with economic implications.

Local → Systemic Amplification: Local impacts (community capacity, cultural preservation, economic transformation) amplify into systemic change (policy reform, institutional transformation) through strategic engagement. Stop LAPD Spying Coalition provides substantiation for this: community capacity to document surveillance (IM2) enables advocacy that produces policy victories (IM1).